

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400253
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Hasan; Ricky

Interactive sales funnel system and method

Abstract

An interactive sales funnel system and method that provides for delivery, execution, management, and optimization of an interactive sales funnel associated with the offering for sale of a product and/or service to a third party. The interactive sales funnel system employs a graphical interface for display on a display device comprising a product need visualization wheel together with a visual questionnaire that are utilized by a consumer to perform a customized product needs assessment. Based on that customized product needs assessment, the graphical interface is updated to show a visual depiction of a relationship between respective consumer inputs, as a function of specific product need assessment categories, received in relation to each other. The consumer then makes an informed purchasing decision based on the depicted visual relationship.

Inventors:	Hasan; Ricky (Troy, FL)
Applicant:	Hasan; Ricky (Troy, FL)
Family ID:	1000006406949
Appl. No.:	17/826890
Filed:	May 27, 2022

Related U.S. Application Data

us-provisional-application US 63194945 20210529

Publication Classification

Int. Cl.: G06Q30/00 (20230101); G06Q30/0601 (20230101)

U.S. Cl.:

CPC G06Q30/0625 (20130101);

Field of Classification Search

CPC: G06Q (30/00); G06Q (30/0601); G06Q (30/0625); G06Q (30/0633); G06Q (30/0641)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6064982	12/1999	Puri	N/A	N/A
6731788	12/2003	Agnihotri	382/165	G11B 27/28
7546248	12/2008	Ellison et al.	N/A	N/A
8200524	12/2011	Narvaez et al.	N/A	N/A
8214238	12/2011	Fairfield et al.	N/A	N/A
8271355	12/2011	Johnson et al.	N/A	N/A
8458053	12/2012	Buron	705/26.1	H04N 21/4316
8660869	12/2013	MacIntyre et al.	N/A	N/A
8660912	12/2013	Dandekar	705/26.1	G06Q 30/02
8660953	12/2013	Enand et al.	N/A	N/A
8977554	12/2014	Hertschuh	704/270	G06Q 30/0623
9972014	12/2017	McCord et al.	N/A	N/A
10068029	12/2017	Lu et al.	N/A	N/A
10270912	12/2018	Piaggio et al.	N/A	N/A
10346621	12/2018	Athuluru Tlrumala et al.	N/A	N/A
10657549	12/2019	Meyer et al.	N/A	N/A
10664534	12/2019	Surya et al.	N/A	N/A
11106751	12/2020	Kadayam	N/A	G06F 16/954
2002/0194055	12/2001	Takakura et al.	N/A	N/A
2003/0129574	12/2002	Ferriol	434/323	G09B 5/00
2003/0167198	12/2002	Northcot et al.	N/A	N/A
2004/0115605	12/2003	Cooper	600/300	G09B 23/28
2005/0108094	12/2004	Hugron	705/14.44	G06Q 30/02
2007/0050406	12/2006	Byers	N/A	G06F 16/48
2007/0106570	12/2006	Hartman	705/26.8	G06Q 30/0605
2007/0112647	12/2006	Borders	705/26.81	G06Q 30/016
2007/0294240	12/2006	Steele	707/999.005	G06F 16/338
2008/0103876	12/2007	Armstrong et al.	N/A	N/A
2008/0103913	12/2007	Leach et al.	N/A	N/A
2008/0162684	12/2007	Neyama	709/223	H04L 67/02
2008/0288889	12/2007	Hunet et al.	N/A	N/A
2008/0294996	12/2007	Hunt et al.	N/A	N/A
2009/0037291	12/2008	Dawson	705/27.2	G06Q 30/0603
2009/0077672	12/2008	Gold	715/202	G06T 9/001
2010/0030578	12/2009	Siddique	705/26.1	G06Q 20/12
2010/0114654	12/2009	Lukose	705/14.54	G06Q 30/00
2011/0078036	12/2010	Cummings	705/1.1	G06Q 10/10
2011/0106581	12/2010	Rohrasser et al.	N/A	N/A
2012/0130817	12/2011	Bousaleh et al.	N/A	N/A

2012/0136756	12/2011	Jitkoff	715/224	G06F 16/9577
2012/0197700	12/2011	Kalin	705/26.8	G06Q 30/0633
2012/0233020	12/2011	Eberstadt	705/26.1	G06Q 30/06
2013/0013427	12/2012	Gonsalves	705/27.1	G06Q 30/02
2013/0097018	12/2012	Davidson et al.	N/A	N/A
2013/0124259	12/2012	Chourey et al.	N/A	N/A
2013/0215116	12/2012	Siddique	705/26.7	G06Q 20/40
2014/0129393	12/2013	Soon-Shiong	705/27.2	G06F 3/0484
2014/0136334	12/2013	Lagassey	705/14.69	G06Q 30/0277
2014/0136990	12/2013	Gonnen	715/752	H04L 51/18
2014/0237394	12/2013	Park	715/758	H04L 51/06
2014/0297362	12/2013	Kumar	705/7.29	G06Q 30/0253
2014/0297537	12/2013	Kassemi	705/67	G06Q 30/0279
2014/0372953	12/2013	Laurance	715/835	G06F 3/04817
2015/0052014	12/2014	Mason	705/26.41	G06Q 30/0643
2015/0052061	12/2014	Anderson	705/44	G06Q 20/409
2015/0088686	12/2014	Glassberg	705/26.8	G06Q 50/01
2015/0149168	12/2014	Stent	704/235	G10L 15/22
2015/0170256	12/2014	Pettyjohn	705/26.9	G06Q 30/0603
2015/0220984	12/2014	Woodget et al.	N/A	N/A
2015/0317698	12/2014	Kalyvas	705/14.66	G06Q 30/0214
2016/0071134	12/2015	Duncan	N/A	N/A
2016/0189073	12/2015	Godewyn et al.	N/A	N/A
2016/0364734	12/2015	Glanville	N/A	N/A
2016/0371854	12/2015	Gershon	N/A	G06Q 30/0643
2017/0032417	12/2016	Amendjian	N/A	G06Q 30/0254
2017/0193434	12/2016	Shah	N/A	G05D 1/0212
2017/0199648	12/2016	Raffo	N/A	G06F 3/04842
2017/0200289	12/2016	Gershon	N/A	G06V 10/751
2019/0340624	12/2018	Kinkead et al.	N/A	N/A
2019/0378200	12/2018	Costello	N/A	G06Q 30/0635
2020/0160938	12/2019	Majumdar	N/A	G06F 16/26
2020/0293568	12/2019	Lu et al.	N/A	N/A
2020/0302510	12/2019	Chachek	N/A	G06V 20/52
2022/0351266	12/2021	Ren	N/A	G06Q 10/06395
2022/0357821	12/2021	Roberts	N/A	G16H 50/70

OTHER PUBLICATIONS

Ramanujan, Devarajan, William Z. Bernstein, and Karthik Ramani. “Design patterns for visualization-based tools in sustainable product design.” International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. vol. 58165. (Year: 2017). cited by examiner

Rosa, Jose Antonio, and Joseph F. Porac. “Categorization bases and their influence on product category knowledge structures.” Psychology & Marketing 19.6 (2002): 503-531. (Year: 2002). cited by examiner

Primary Examiner: Seibert; Christopher B

Attorney, Agent or Firm: John Rizvi, P.A.—The Patent Professor®

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application 63/194,945, filed May 29, 2021, which is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to sales funnel operations, and more particularly, to a system and method that facilitate an interactive sales funnel experience between a consumer and a salesperson that allow the consumer to interactively define and assess their individual needs for a particular product or service.

BACKGROUND OF THE INVENTION

(2) The marketing and selling of products and services is a cornerstone in any thriving economy. The buying and selling of products and services has resulted in a vast array of schemes which are used to vary the price at which such products are sold. The so-called purchase funnel, or purchasing funnel, is a consumer-focused marketing model that illustrates the theoretical customer journey toward the purchase of a good or service. The so-called “AIDA-model” is a model that maps a theoretical customer journey from the moment a brand or product attracted consumer attention to the point of action or purchase. The AIDA acronym stands for Awareness, Interest, Desire, and Action. This staged process is summarized as: (1) awareness—the customer is aware of the existence of a product or service; (2) interest—actively expressing an interest in a product group; (3) desire—aspiring to a particular brand or product, and (4) action—taking the next step towards purchasing the chosen product. The purchase funnel is alternatively referred to as the “customer funnel”, “marketing funnel” or the “sales funnel”. Some variants of the AIDA-model replace attention with awareness. The common thread among all hierarchical models is that advertising operates as a stimulus and the purchase decision is a response. In other words, the AIDA-model is an applied stimulus-response model.

(3) The basic AIDA-model is one of the longest serving hierarchical models, having been in use for more than a century. Using a hierarchical system, such as AIDA, provides the marketer with a detailed understanding of how target audiences change over time, and provides insights as to which types of advertising messages are likely to be more effective at different junctures. Moving from step to step, the total number of prospects diminishes. This phenomenon is sometimes described as a “purchase funnel”. A relatively large number of potential purchasers become aware of a product or brand, then a smaller subset becomes interested, with only a relatively small proportion moving through to the actual purchase. This effect is also known as a “customer funnel”, “marketing funnel”, or “sales funnel”. As consumers move through the hierarchy of effects, they pass through both a cognitive processing stage and an affective processing stage before any action occurs. Thus, the hierarchy of effects models all include cognition-affect-behavior as the core steps in the underlying behavioral sequence. As such, the sales funnel is a step-by-step process that allows a salesperson to bring a potential customer one step closer to a product offer and a buying decision through a series of marketing actions such as automated emails, videos, articles, and landing pages that assist in the selling process. For example, a landing page is a page that should clearly communicate who you are as a company and your unique benefits that can benefit the potential customer and collect information from the potential customer so the salesperson can continue to communicate with them at a later time.

(4) In today's electronic commerce world and with the expansion of the Internet and the increase of access to the Internet and other electronic commerce vehicles, a wide variety of web-based interfaces and systems have been developed focused on commerce. These interfaces and systems often enhance the value and utility provided by an Internet web site or electronic information

system or service. The success of an Internet web site, for example, may strongly depend on the quality of the utilities provided by the web site. In addition, the growth in the field of Internet web sites has demanded that competitive companies provide more than a single resource on their web site. For example, customer relationship management (CRM) systems and methods are used by organizations to provide a predictable and organized way for interacting with existing customers and potential customers. CRM typically includes specially trained personnel and special purpose software. It is a combination of policies, processes and strategies implemented by an organization to unify customer interactions and provide a method for tracking customer information. CRM often also includes technology for identifying and attracting new and profitable customers as well as creating better relationships with existing customers. CRM involves many organizational aspects that relate to one another, including front and back-office operations, business relationships and interactions, analysis involving target marketing and marketing strategies, and means for generating metrics for measuring the relative success of various marketing and sales efforts. It is one key component of modern marketing organizations and CRM systems include firmographic data, which includes characteristics of an organization often used for segment market analysis.

(5) Similarly, marketing automation systems and methods are used by organizations to communicate with prospects and customers and automate many marketing communication tasks. Whereas a CRM system is often leveraged as a database for the sales organization, a marketing automation solution is mostly leveraged as a database for the marketing organization. Marketing automation is often leveraged to communicate with customers and prospects via email, track and report on campaign responses, profile the quality and sales-readiness of leads generated by marketing programs, prioritize which leads are passed to members of a sales team, and to automate ongoing communication.

(6) In addition to the individual consumer-seller retail relationship, the business-to-business (B2B) marketing relationship includes individuals and organizations that facilitate the sale of their products and services to other companies or organizations that often resell the products and services or use them to support their own operations. Although the difference between consumer-centric and business-centric marketing may appear obvious, there are many distinguishing features between the two that often result in substantial differences in practice. For example, B2B marketing may often involve shorter and more direct channels of distribution. While consumer marketing often involves large demographic groups targeted through mass media and retailers, in B2B marketing the negotiation process between the seller and buyer may be more direct/personal in nature. B2B marketing and sales is conducted through more direct promotional efforts, trade journals and sales calls. However, many of the principles of consumer marketing also apply to B2B marketing, such as defining target markets and matching product and service strengths to the defined target markets. One common and fundamental principle prevails in both worlds, namely, successful sales in high volumes and profitability. The ultimate goal of any of the above sales and/or marketing techniques is to make high volume sales, with the least amount of wasted effort and at the largest profit.

(7) While the sales and marketing schemes described above have various advantages and disadvantages in different situations, a commonality among all of the buying schemes is that each buyer is not given the opportunity to choose amongst different buying criteria, which could be more important to the buyer than the price of the goods and/or services. Furthermore, each seller is not given a chance to offer their goods and/or services based on different selling criteria. In many instances sellers may not even be aware of what buyers consider important buying criteria. Additionally, sellers are sometimes not aware of what other selling criteria that other sellers might offer buyers. As such, a great deal of effort goes into analyzing marketing data and optimizing the sales and marketing processes and product sales funnel.

(8) Accordingly, there is need for a technique that improves sales funnel efficiency and the acceleration of conversions through the sales funnel to facilitate a positive real-time purchasing

decision.

SUMMARY OF THE INVENTION

(9) The present invention is directed to an interactive sales funnel system and method that provides for delivery, execution, management, and optimization of an interactive sales funnel associated with the offering for sale of a product and/or service to a third party. The interactive sales funnel system employs a graphical interface for display on a display device comprising a product need visualization wheel together with a visual questionnaire that are utilized by a consumer to perform a customized product needs assessment. Based on that customized product needs assessment, the graphical interface is updated to show a visual depiction of a relationship between respective consumer inputs, as a function of specific product need assessment categories, received in relation to each other. The consumer then makes an informed purchasing decision based on at least the depicted visual relationship.

(10) In a first implementation of the invention, an interactive sales funnel system is provided comprising a display device, a processor; a memory storing instructions that when executed cause the processor to perform operations comprising: (i) generating, by the processor, a graphical interface for display on the display device, the graphical interface comprising a product need visualization wheel together with a visual questionnaire, the product need visualization wheel comprising a form of a circular wheel with a center point, a circumference and a plurality of spokes, each spoke configured between the center point and a particular one point along the circumference of the wheel in a symmetric configuration thereabout, the visual questionnaire comprising a plurality of individual product need assessment categories specific to a product offered for sale to a consumer in accordance with criteria associated with a sales funnel, each individual product need assessment category of the plurality of product need assessment categories having a corresponding graduated consumer assessment scale associated therewith, the graduated consumer assessment scale comprising a plurality of metrics for establishing the consumer's need for the product offered for sale, each particular one spoke of the plurality of spokes corresponding with a particular one individual product need assessment category of the plurality of product need assessment categories, and the product need visualization wheel showing, along a respective one spoke of the plurality of spokes, a respective one graduated consumer assessment scale associated with each individual product need assessment category of the plurality of individual product need assessment categories; (ii) displaying, on the display device, the graphical interface generated; (iii) receiving, from the consumer, a plurality of inputs as a function of the plurality of individual product need assessment categories displayed by the graphical interface and the corresponding graduated consumer assessment scale associated therewith, each input of the plurality of inputs received being specific to a respective one individual product need assessment category of the plurality of individual product need assessment categories and comprising at least a selection of a respective one metric of the plurality of metrics from the graduated consumer assessment scale; (iv) updating, by the processor, the graphical interface displayed as function of the plurality of inputs received such that the graphical interface updated shows a respective one graduated consumer assessment scale associated with each individual category of a plurality of individual categories together with each input of the plurality of inputs received and the respective one metric of the plurality of metrics from the graduated consumer assessment scale selected, and generating, as part of the graphical interface updated, a visual depiction of a relationship between each respective one input received in relation to each other; (v) displaying the graphical interface updated; and (vi) receiving, from the consumer a selection, as a function of the graphical interface updated and the visual depiction of the relationship between each respective one input received in relation to each other, for the product offered for sale.

(11) In a second aspect, a method is provided for managing an interactive sales funnel comprising operations of: (i) generating, by the processor, a graphical interface for display on the display device, the graphical interface comprising a product need visualization wheel together with a visual

questionnaire, the product need visualization wheel comprising a form of a circular wheel with a center point, a circumference and a plurality of spokes, each spoke configured between the center point and a particular one point along the circumference of the wheel in a symmetric configuration thereabout, the visual questionnaire comprising a plurality of individual product need assessment categories specific to a product offered for sale to a consumer in accordance with criteria associated with a sales funnel, each individual product need assessment category of the plurality of product need assessment categories having a corresponding graduated consumer assessment scale associated therewith, the graduated consumer assessment scale comprising a plurality of metrics for establishing the consumer's need for the product offered for sale, each particular one spoke of the plurality of spokes corresponding with a particular one individual product need assessment category of the plurality of product need assessment categories, and the product need visualization wheel showing, along a respective one spoke of the plurality of spokes, a respective one graduated consumer assessment scale associated with each individual product need assessment category of the plurality of individual product need assessment categories; (ii) displaying, on the display device, the graphical interface generated; (iii) receiving, from the consumer, a plurality of inputs as a function of the plurality of individual product need assessment categories displayed by the graphical interface and the corresponding graduated consumer assessment scale associated therewith, each input of the plurality of inputs received being specific to a respective one individual product need assessment category of the plurality of individual product need assessment categories and comprising at least a selection of a respective one metric of the plurality of metrics from the graduated consumer assessment scale; (iv) updating, by the processor, the graphical interface displayed as function of the plurality of inputs received such that the graphical interface updated shows a respective one graduated consumer assessment scale associated with each individual category of a plurality of individual categories together with each input of the plurality of inputs received and the respective one metric of the plurality of metrics from the graduated consumer assessment scale selected, and generating, as part of the graphical interface updated, a visual depiction of a relationship between each respective one input received in relation to each other; (v) displaying the graphical interface updated; and (vi) receiving, from the consumer a selection, as a function of the graphical interface updated and the visual depiction of the relationship between each respective one input received in relation to each other, for the product offered for sale.

(12) In a third aspect, an interactive sales funnel system application (alternatively referred to herein as an “app”) for executing, on a user device or other hardware, is provided comprising operations of: (i) generating, by the processor, a graphical interface for display on the display device, the graphical interface comprising a product need visualization wheel together with a visual questionnaire, the product need visualization wheel comprising a form of a circular wheel with a center point, a circumference and a plurality of spokes, each spoke configured between the center point and a particular one point along the circumference of the wheel in a symmetric configuration thereabout, the visual questionnaire comprising a plurality of individual product need assessment categories specific to a product offered for sale to a consumer in accordance with criteria associated with a sales funnel, each individual product need assessment category of the plurality of product need assessment categories having a corresponding graduated consumer assessment scale associated therewith, the graduated consumer assessment scale comprising a plurality of metrics for establishing the consumer's need for the product offered for sale, each particular one spoke of the plurality of spokes corresponding with a particular one individual product need assessment category of the plurality of product need assessment categories, and the product need visualization wheel showing, along a respective one spoke of the plurality of spokes, a respective one graduated consumer assessment scale associated with each individual product need assessment category of the plurality of individual product need assessment categories; (ii) displaying, on the display device, the graphical interface generated; (iii) receiving, from the consumer, a plurality of inputs as a function of the plurality of individual product need assessment categories displayed by the

graphical interface and the corresponding graduated consumer assessment scale associated therewith, each input of the plurality of inputs received being specific to a respective one individual product need assessment category of the plurality of individual product need assessment categories and comprising at least a selection of a respective one metric of the plurality of metrics from the graduated consumer assessment scale; (iv) updating, by the processor, the graphical interface displayed as function of the plurality of inputs received such that the graphical interface updated shows a respective one graduated consumer assessment scale associated with each individual category of a plurality of individual categories together with each input of the plurality of inputs received and the respective one metric of the plurality of metrics from the graduated consumer assessment scale selected, and generating, as part of the graphical interface updated, a visual depiction of a relationship between each respective one input received in relation to each other; (v) displaying the graphical interface updated; and (vi) receiving, from the consumer a selection, as a function of the graphical interface updated and the visual depiction of the relationship between each respective one input received in relation to each other, for the product offered for sale.

(13) In a fourth aspect, the interactive sale funnel app may be a mobile application executing on a mobile device and wherein the mobile device may be a smartphone, laptop computer, tablet and/or wearable device.

(14) In a fifth aspect, the product offered for sale is related to toll management, and the plurality of individual product need assessment categories include violations, stress level, consolidation, need to manage, number of transponders, discounts, managing system, and time allocated.

(15) In a sixth aspect, a uniformity of the visual depiction of the relationship between each respective one input received in relation to each other is used to determine a consumer's product needs.

(16) In a seventh aspect, the graduated consumer assessment scale is a numerical scale.

(17) In another aspect, the numerical scale comprises a range from 1 to 10, wherein an assessment of "1" represents a worst or lowest rating possible rating and an assessment of "10" represents a best or highest rating possible.

(18) In another aspect, the greater the uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other represents a lesser need for the product offered for sale by the consumer.

(19) In another aspect, the lesser the uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other represents a greater need for the product offered for sale by the consumer.

(20) In another aspect, if the visual depiction generated of the relationship between each respective one input received in relation to each other in any one visual depiction is that of a circle, then the relationship between the plurality of metrics, as depicted by the visual depiction generated, is indicative of a lesser need for the product offered for sale by the consumer.

(21) In another aspect, the product need visualization wheel and the visual questionnaire are fully customizable.

(22) These and other objects, features, and advantages of the present invention will become more readily apparent from the attached drawings and the detailed description of the preferred embodiments, which follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The preferred embodiments of the invention will hereinafter be described in conjunction with the appended drawings provided to illustrate and not to limit the invention, where like designations denote like elements, and in which:

- (2) FIG. 1 presents a high-level block diagram of a cloud network service architecture for providing an interactive sales funnel system in accordance with an embodiment.
- (3) FIG. 2 presents an illustrative interactive sales funnel system in accordance with an embodiment.
- (4) FIG. 3 presents an illustrative user device configured for use with the interactive sales funnel system of FIG. 2 in accordance with an embodiment.
- (5) FIG. 4 presents an illustrative architecture for an interactive sales funnel system app in accordance with an embodiment.
- (6) FIG. 5 presents a flowchart of illustrative operations for providing an interactive sales funnel for use with the interactive sales funnel system of FIG. 2 in accordance with an embodiment.
- (7) FIGS. 6, 7, 8 and 9 present a series of illustrative graphical interfaces generated by the interactive sales funnel system of FIG. 2 in accordance with an embodiment.
- (8) Like reference numerals refer to like parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(9) The following detailed description is merely exemplary in nature and is not intended to limit the described embodiments or the application and uses of the described embodiments. As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. For purposes of description herein, the terms “upper”, “lower”, “left”, “rear”, “right”, “front”, “vertical”, “horizontal”, and derivatives thereof shall relate to the invention as oriented in the figures herein. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(10) Shown throughout the figures, the present invention is directed toward an interactive sales funnel system and method that provides for delivery, execution, management, and optimization of an interactive sales funnel associated with the offering for sale of a product and/or service to a third party (e.g., individual consumer or B2B purchaser). Importantly, the interactive sale funnel system of the disclosed embodiments provides an advantageous improvement of at least one practical application, i.e., the product sales application, in that the individual consumer will play a direct and interactive role in the engagement with the salesperson. In accordance with the disclosed embodiments, that practical sales application provides for an interactive engagement, by and through a specifically configured application executing on a hardware system, with the individual consumer that allows the consumer to play an important role in the overall sales process and coming to their own realization of whether they need a particular product or not based on a customized needs assessment. In this way, the customer is actively engaged and formulates their own product need assessment, for example, through the AIA-model steps of (1) awareness—the customer is aware of the existence of a product or service; (2) interest—actively expressing an interest in a product group; (3) desire—aspiring to a particular brand or product; and (4) action—taking the next step towards purchasing the chosen product. This interactive engagement, in accordance with the disclosed embodiments, is facilitated by a customized visualization tool and graphical interface, as further detailed herein below. The interactive sales funnel system employs the graphical interface for display on a display device comprising a product need visualization wheel together with a visual questionnaire that are utilized by a consumer to perform a customized

product needs assessment. Based on that customized product needs assessment, the graphical interface is updated to show a visual depiction of a relationship between respective consumer inputs, as a function of specific product need assessment categories, received in relation to each other. The consumer then makes an informed purchasing decision based on at least the depicted visual relationship.

(11) FIG. 1 presents a high-level block diagram of a cloud network service architecture **100** for delivering, executing, managing, and optimizing an interactive sales funnel associated with the offering for sale of a product and/or service to a third party (e.g., individual consumer or B2B purchaser) in accordance with an embodiment. For clarity and the avoidance of doubt, it will be understood that the use of the term “consumer” herein means any type of consumer including, but not limited, an individual consumer and B2B purchaser. Similarly, the use of the term “product” herein shall mean both a product and/or service that is offered for sale to a consumer. As shown for instance in FIG. 1, the cloud network service architecture **100** includes a cloud **102** comprising at least server(s) **104**, access point(s) **106** and database(s) **108**. As will be detailed herein below, the cloud **102** facilitates the delivery of the interactive sales funnel using interactive sales funnel system **200** to a plurality of consumers (e.g., the plurality consumers comprised by consumer 1 **110-1** through consumer N **110-N**) and a plurality of sellers (e.g., seller 1 **118-1** through seller N **118-N**), whereby the individual consumers will play an important role in the overall sales process and coming their own realization of whether they need a particular product or not based on a customized needs assessment made through their use a customized visualization tool and graphical interface as provided in accordance with the principles of the disclosed embodiments. In an embodiment, the customized visualization tool and graphical interface, and associated operations, offered by and through the cloud network service architecture **100** will be facilitated by an interactive sales funnel app (see, FIG. 4), as will be detailed herein below, executing on a user device **300** (see, FIG. 3). The user device **300** provides consumers (e.g., consumer 1 **110-1** through consumer N **110-N**) with real-time access to sales funnel services for the purchasing of products in accordance with the disclosed embodiments herein.

(12) As noted above, the cloud **102** comprises at least server(s) **104**, the access point(s) **106** and the database(s) **108**. Cloud, cloud service, cloud server and cloud database are broad terms and are to be given their ordinary and customary meaning to one of ordinary skill in the art and includes, without limitation, any database, data repository or storage media which store content typically associated with and managed by users, product sales and marketing services, product sales funnel providers, third party content providers, social media services, to name just a few. A cloud service may include one or more cloud servers and cloud databases that provides for the remote storage of content as hosted by a third-party service provider or operator. A cloud server may include an HTTP/HTTPS server sending and receiving messages in order to provide web-browsing interfaces to client web browsers as well as web services to send data to integrate with other interfaces (e.g., as executed on the user device **300**). The cloud server may be implemented in one or more servers and may send and receive content in a various forms and formats, user supplied and/or created information/content and profile/configuration data that may be transferred to, read from or stored in a cloud database (e.g., the databases **108**). The cloud **102** may also facilitate the use by one or more seller of the plurality of sellers, advertisers **142**, social media services **120** and/or third-party content providers **114** in the course of the product offerings and the use of the interactive sales funnel system **200**.

(13) A cloud database may include one or more physical servers, databases or storage devices as dictated by the cloud service's storage requirements. The cloud database may further include one or more well-known databases (e.g., an SQL database) or a fixed content storage system to store content, profile information, configuration information or administration information as necessary to execute the cloud service. In various embodiments, one or more networks providing computing infrastructure on behalf of one or more users may be referred to as a cloud, and resources may

include, without limitation, data center resources, applications (e.g., software-as-a-service or platform-as-a-service) and management tools. In this way, in accordance with various embodiments, the users may control, initiate, and engage in the interactive sales funnel services herein in a fully transparent fashion without any required understanding of the underlying hardware and software necessary to interface, communicate, manipulate, and exchange information and/or data necessary to deliver such services.

(14) Turning our attention to FIG. 2, an illustrative configuration for the interactive sales funnel system **200** is shown for deployment in the cloud network service architecture **100** in accordance with an embodiment. As shown, the interactive sale funnel system **200** comprises processor **202** for executing program code (e.g., interactive sales funnel system app **400**) and communications interface **214** for managing communications to and from the interactive sales funnel system **200**, memory **206** and/or ROM **208** for storing program code and data, and power source **218** for powering the interactive sale funnel system **200**. The memory **206** is coupled to the bus **204** for storing computer-readable instructions to be executed by the processor **200** (e.g., execution of the interactive sales funnel system app **400**). Database manager **212** is used to manage the delivery and storage of content, data, and other information in the interactive sales funnel database(s) **124** and across third party content providers, for example. The interactive sales funnel database(s) **124** may store and provide information including, but not limited, to consumer IDs **130**, Seller IDs **132**, product information **134**, product need assessment categories **136**, sales funnel information **138**, and consumer assessment scale metrics **140**. Similarly, website manager **220** is used to deliver and manage content, data, and other information across one or more websites that may be utilized to access and use the interactive sales funnel system **200**, for example. As will be discussed in greater detail herein below, the interactive sales funnel system app **400**, as stored in data storage **210**, when executed by the processor **202** will enable access by the consumer (e.g., consumer 1 **110-1**) to the interactive sales funnel system **200** for delivery, execution, management, and optimization of an interactive sales funnel associated with the offering for sale of a product and/or service to a third-party (e.g., individual consumer or B2B purchaser). Location-based service manager **222** facilitates the delivery of location-based services (e.g., GPS tracking) either independently or on user device **300** thereby allowing the interactive sales funnel system **200** to register the exact location of the user of the user device **300**, for example, as the user roams from one location to another location such that the products offered via the interactive sales funnel may be tailored to a current location and/or the product needs of the user/consumer may change based on their current location. Importantly, the interactive sale funnel system **200** provides an advantageous improvement of at least one practical application, i.e., the product sales application, in that the individual consumer will play a direct and interactive role in the engagement with the salesperson. In accordance with the disclosed embodiments, that practical sales application provides for an interactive engagement, by and through a specifically configured application executing on a hardware system, with the individual consumer will allow the consumer to play an important role in the overall sales process and coming to their own realization of whether they need a particular product or not based on a customized needs assessment. This interactive engagement with the consumer, in accordance with the disclosed embodiments, is facilitated by a customized visualization tool and graphical interface, as further detailed herein below, that is displayed by and through display device **224**, for example, using graphical interface generator **226**. It will be understood that while the instant embodiment shows the display device **224** integrated within the interactive sales funnel system **200** this is not limiting in nature and the display device **224** may be independent of and external to the interactive sales funnel system **200** in accordance with principles of the disclosed embodiments herein.

(15) In a further embodiment, the interactive sales funnel services provided through the execution of the interactive sales funnel system app **400** may also include accessing and interfacing any number of web pages (e.g., as constructed and hosted by the third-party content providers **114**) with a variety of third-party websites using website manager **220** for procuring information and data that

can be used in the interactive sales funnel system. The term “website” in the context herein is used in a conventional and broadest sense and is located on at least one server containing web pages stored thereon and is operational in a 24-hour/7-day typical fashion. As shown in the interactive sales funnel system **200**, the plurality of consumers (i.e., consumer 1 **110-1** through consumer N **110-N**) and/or the plurality of sellers (i.e., seller 1 **118-1** through seller N **118-N**) may alternatively utilize well-known Internet **122** for access to interactive sales funnel system **200** by and through the user device **300**.

(16) The communications interface **214** is used to facilitate communications across the communications links **128** (see, FIG. **1**) within the cloud **102** in a well understand fashion. This may take the form, for example, of a wide area network connection that communicatively couples the interactive sales funnel system **200** with the access points **106** (see, FIG. **1**) which may be a cellular communications service. Similarly, communications managed by the communications interface **214** may take the form, for example, of a local Wi-Fi network interface or Ethernet interface the communicatively couples the interactive sale funnel system **200** with the Internet **122**, LAN **112** and ultimately the user device **300**. In the instant embodiment, the interactive sales funnel system app **400** and/or the communications interface **214** may include a communications stack for facilitating communications over the respective communications link **128**. Electronic communications by and through interactive sales funnel system **200** between the various systems, networks, devices, users, entities, and/or individuals are facilitated by the communications links **128** in accordance with any number of well-known communications protocols and methods (e.g., wireless communications).

(17) Turning our attention briefly to FIG. **3**, an illustrative user device **300** is shown for deployment with the cloud network service architecture **100** of FIG. **1** in accordance with an embodiment. The user device **300** includes bus **302** and processor **304** coupled to the bus **302** for executing operations and processing information. As will be appreciated, a “user device” in the context herein may comprise a wide variety of devices such as any type of mobile device, smartphones, laptop computers, tablets, and wearable device, to name just a few, that execute applications (e.g., a mobile application) in accordance with the principles of the disclosed embodiments herein. For example, the execution of the operations of interactive sales funnel system app **400** as will be discussed in much greater detail herein below. The processor **304**, as powered by power source **312**, may include both general and special purpose microprocessors, and may be the sole processor or one of multiple processors of the device. This holds true also for the processor **202** of FIG. **2**. Further, the processor **304** (or the processor **202**) may comprise one or more central processing units (CPUs) and may include, be supplemented by, or incorporated in, one or more application-specific integrated circuits (ASICs) and/or one or more field programmable gate arrays (FPGAs).

(18) The user device **300** may also include main memory **306** coupled to the bus **302** for storing computer-readable instructions to be executed by the processor **304** in a well-known manner. The memory **306** may also be utilized for storing temporary variables or other intermediate information during the execution of the instructions by the processor **304**. The user device **300** may also include read-only memory (ROM) **308** or other static storage device coupled to the bus **302**. Further, data storage device **310**, such as a magnetic, optical, or solid-state device may be coupled to the bus **302** for storing information and instructions for the processor **304** including, but not limited to, the interactive sales funnel system app **400**. Data storage device **310** (or the data storage device **210**) and the memory **306** (or the memory **206**) may each comprise a tangible non-transitory computer readable storage medium and may each include high-speed random access memory, such as dynamic random access memory (DRAM), static random access memory (SRAM), double data rate synchronous dynamic random access memory (DDR RAM), or other random access solid state memory devices, and may include non-volatile memory, such as one or more magnetic disk storage devices such as internal hard disks and removable disks, magneto-optical disk storage devices, optical disk storage devices, flash memory devices, semiconductor memory devices, such as

erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), compact disc read-only memory (CD-ROM), digital versatile disc read-only memory (DVD-ROM) disks, or other non-volatile solid state storage devices.

(19) The user device **300** may also include one or more communications interface **316** for communicating with other devices via a network (e.g., a wireless communications network) or communications protocol (e.g., Bluetooth®). For example, such communication interfaces may be a receiver, transceiver, or modem for exchanging wired or wireless communications in any number of well-known fashions. For example, the communications interface **316** (or the communications interface **214**) may be an integrated services digital network (ISDN) card or modem/router used to facilitate data communications of various well-known types and formats. Further, illustratively, the communications interface **316** (or the communications interface **214**) may be a local area network (LAN) card used to provide data communication connectivity to a comparable LAN. Wireless communication links may also be implemented. The Global Positioning System (GPS) transceiver **318** and antenna **320** facilitate delivery of location-based services in a well-known manner to register the exact location of the user of the user device **300**, for example, as the user roams from one location to another location the products offered via the interactive sales funnel may be tailored to a current location and/or the product needs of the user/consumer may change based on their current location.

(20) As will be appreciated, the functionality of the communication interface **316** (or the communications interface **214**) is to send and receive a variety of signals (e.g., electrical, optical, or other signals) that transmit data streams representing various data types. The user device **300** may also include one or more input/output devices **314** that enable user interaction with the user device **300** (e.g., camera, display, keyboard, mouse, speakers, microphone, buttons, etc.). The input/output devices **314** (or the I/O devices **216**) may include peripherals, such as a camera, printer, scanner, display screen, etc. For example, the input/output devices **314** (or I/O devices **214**) may include a display device such as a cathode ray tube (CRT), plasma monitor, liquid crystal display (LCD) monitor or organic light-emitting diode (OLED) monitor for displaying information to the user, a keyboard, and a pointing device such as a mouse or a trackball by which the user can provide input to the user device **300** or an associated display device, for example. As detailed herein, while FIG. **3** describes an embodiment of the interactive sales funnel system app **400** for execution, illustratively, on the user device **300** it will also be understood that other hardware devices may be used to execute and operate the interactive sales funnel system app **400** in any real-time setting including but not limited to the interactive sales funnel system **200**. For example, a network-enabled portable tablet computer and/or dedicated portable hardware device may be employed equally in the context of the disclosed embodiments.

(21) Turning our attention to FIG. **4**, an illustrative architecture for the operation of the interactive sales funnel system app **400** is presented in accordance with an embodiment. As will be appreciated, the architecture may be used in conjunction with the interactive sales funnel system **200** and/or the user device **300** for launching and executing the interactive sales funnel system app **400** and its associated operations. As shown, the architecture for the operation of the interactive sales funnel system app **400** provides several interfaces and engines used to perform a variety of functions such as the collection, aggregation, manipulation, processing, analyzing and display of applicable real-time information and data that are useful to achieve the interactive sales funnel services of the disclosed embodiments. More particularly, data display interface module **418** and communications module **412** are used to facilitate the input/output and display of electronic data and other information (e.g., a graphical user interface) to, illustratively, the users (e.g., consumer 1 **110-1** through consumer N **110-N**) employing the user device **300** (e.g., a touch screen of the user device **300**) and executing the interactive sales funnel system app **400**. The data collection interface module **406** facilitates the collection of sales funnel information from the plurality of consumers (i.e., consumer 1 **110-1** through consumer N **110-N**), the plurality of sellers (i.e., seller 1 **118-1**

through seller N **118-N**) and/or other third parties. The data display interface module **418** facilitates the delivery of the graphical interface that enables the interactive sales funnel services, as will be further detailed herein below. The location-based services module **420** provides for the delivery of location-based services in order for the geographic locations of the consumers to be identified and displayed (e.g., GPS locations). The communications module **412** will facilitate communications by and through the interactive sales funnel system **200**, for example.

(22) Execution engine **402** may be employed to deliver the interactive sales funnel services herein through the execution of the interactive sales funnel system app **400**. In such delivery, the execution engine **402** will operate and execute, as further detailed herein below, with at least the following program modules: graphical interface module **404**, data collection module **406**, data display interface module **418**, location-based services module **420**, product need assessment module **408**, visual questionnaire module **410**, communications module **412**, interactive sales funnel operations module **414**, and consumer assessment and metrics module **416**. More particularly, the graphical interface module **404**, the data collection module **406**, the data display interface module **418**, the product need assessment module **408**, the visual questionnaire module **410**, the communications module **412**, the interactive sales funnel operations module **414**, and the consumer assessment and metrics module **416** will provide for operations that when executed will deliver features of generating a unique graphical interface for display on a display device. As will be detailed further herein below, the interactive sales funnel services delivered herein, as managed by the interactive sales funnel operations module **414**, utilize a graphical interface comprising a product need visualization wheel together with a visual questionnaire, the product need visualization wheel comprising a form of a circular wheel with a center point, a circumference and a plurality of spokes, each spoke configured between the center point and a particular one point along the circumference of the wheel in a symmetric configuration thereabout. The visual questionnaire, delivered and managed by the visual questionnaire module **410**, comprising a plurality of individual product need assessment categories specific to a product offered for sale to a consumer in accordance with criteria associated with a sales funnel, each individual product need assessment category of the plurality of product need assessment categories having a corresponding graduated consumer assessment scale associated therewith, and the graduated consumer assessment scale comprising a plurality of metrics for establishing the consumer's need for the product offered for sale.

(23) In an embodiment, the sales funnel managed by the interactive sales funnel system **200**, for example, is directed to a toll management product that may be used by transportation professionals to manage infrastructure tolls (e.g., highways, tunnels, bridges, and the like) collected by tolling authorities. As such, the plurality of product need assessment categories, as delivered and managed by the product need assessment module **408**, that will be employed in the visual questionnaire include product need assessment categories such as: number of transponders, discounts, violations, consolidation, stress level, need to manage, managing system and time allocated, as will be further discussed herein below. In turn, each of these product need assessment categories will have a graduated consumer assessment, as delivered and managed by the consumer assessment and metrics module **416**, ranging from 1 to 10 that a consumer using the interactive sales funnel system **200** will be presented with for their customized assessment ratings. Illustratively, an assessment of “1” represents the worst or lowest rating possible and an assessment of “10” represents the best or highest rating possible. It will be noted that while the illustrative embodiments herein utilize a numerical scale of 1 to 10 this is only one of a plethora of graduated scales that may be used to form the graduated consumer assessment as used herein. Similarly, while the illustrative embodiments herein utilize a toll management product and associated sales funnel factors this is only one of a plethora of products for which the principles of the embodiments herein apply and the number or type of products can be of any nature depending upon only desires of the sellers and/or the relevant market. That is, the product need visualization wheel and/or the visual questionnaire

herein are fully customizable to meet the needs of any particular product and/or any particular market. With respect to the product need visualization wheel, each particular one spoke of the plurality of spokes corresponds with a particular one individual product need assessment category of the plurality of product need assessment categories, and the product need visualization wheel will show, along a respective one spoke of the plurality of spokes, a respective one graduated consumer assessment scale associated with each individual product need assessment category of the plurality of individual product need assessment categories. The fundamentals of such product need visualization wheel and its associated utility will now be discussed.

(24) Turning our attention to FIG. 6, one of a series of illustrative graphical interfaces generated by the interactive sales funnel system 200 is shown. As shown, graphical interface 600 comprises product need visualization wheel 602 having a plurality of spokes (i.e., spoke 604-1, spoke 604-2, spoke 604-3, spoke 604-4, spoke 604-5, spoke 604-6, spoke 604-7, and spoke 604-8). In turn, each of the spokes has a graduated consumer assessment scale (i.e., graduated consumer assessment scale 606-1, graduated consumer assessment scale 606-2, graduated consumer assessment scale 606-3, graduated consumer assessment scale 606-4, graduated consumer assessment scale 606-5, graduated consumer assessment scale 606-6, graduated consumer assessment scale 606-7, and graduated consumer assessment scale 606-8). It will be noted that while the illustrative embodiments herein utilize a product need visualization wheel having an eight-spoked wheel configuration this is only one of a plethora of configurations that may be used to form the product need visualization wheel as used herein.

(25) In accordance with the embodiment, the product need visualization wheel 602 is in the form of a circular wheel with a center point 608 and a circumference 610 such that the plurality of spokes is rendered with each spoke configured between the center point 608 and a particular one point along the circumference 610 (i.e., point 612-1, point 612-2, point 612-3, point 612-4, point 612-5, point 612-6, point 612-7, and point 612-8) in a symmetric configuration thereabout. In this way, the product need visualization wheel 602 and its constituent components will be rendered on a display device (e.g., the display device 224). Also, each product need assessment category has its own “radius” or “arm” that extends between the center of the wheel and a corresponding equidistant point along the circumference of the wheel. For example, the radius between the center 608 and the point 612-1 that encompasses spoke 604-1 and the graduated consumer assessment scale 606-1. In accordance with the embodiment, a plurality of inputs is received for a consumer having been presented with the product need visualization wheel 602 together with visual questionnaire 616. Such inputs rendered by, and received from, the consumer as a function of the plurality of individual product need assessment categories displayed by the graphical interface using the visual questionnaire 616 and the corresponding graduated consumer assessment scale associated therewith. As shown, the plurality of product need assessment categories associated with the visual questionnaire 616 as displayed are number of transponders 614-1, discounts 614-2, violations 614-3, consolidation 614-4, stress level 614-5, need to manage 614-6, managing system 614-7 and time allocated 614-8.

(26) Turning our attention now to FIG. 7 and graphical interface 601 thereof, in accordance with the embodiment, each input of the plurality of inputs received is specific to a respective one individual product need assessment category of the plurality of individual product need assessment categories. That is, each of the number of transponders 614-1, the discounts 614-2, the violations 614-3, the consolidation 614-4, the stress level 614-5, the need to manage 614-6, the managing system 614-7 and the time allocated 614-8, and said input further comprises at least a selection of a respective one metric of the plurality of metrics from the graduated consumer assessment scale. As shown, the consumer has made input selection 618-1 (i.e., 10 out of 10), input selection 618-2 (i.e., 1 out of 10), input selection 618-3 (i.e., 10 out of 10), input selection 618-4 (i.e., 1 out of 10), input selection 618-5 (i.e., 10 out of 10), input selection 618-6 (i.e., 1 out of 10), input selection 618-7 (i.e., 10 out of 10), and input selection 618-8 (i.e., 1 out of 10). In turn, there is then an updating of

the graphical interface displayed as function of the plurality of inputs received such that the updated graphical interface shows a respective one graduated consumer assessment scale associated with each individual category of a plurality of individual categories together with each input of the plurality of inputs received and the respective one metric of the plurality of metrics from the graduated consumer assessment scale selected, and generating, as part of the graphical interface updated, a visual depiction **620**, as shown in FIG. 7, of a relationship between each respective one input received in relation to each other. In the illustration of FIG. 7, the visual depiction **620** is a star-like shape indicating a serious need of the product under assessment and being offered for sale. That is, in accordance with the principles of the disclosed embodiments, the more uniform in nature that the visual depiction rendered is this will depict a lesser need for the product based on the customer's own assessment.

(27) Said another way, if the desired rendered relationship rendered in any one visual depiction is that of a circle, then this relationship between the plurality of metrics (as determined in accordance with the consumer's direct and interactive product needs assessment inputs) is the most desirable and indicative of a lesser (or no) need for the product being offered for sale. These principles are further illustrated by FIG. 8 and FIG. 9, respectively. In particular, as shown in FIG. 8 and graphical interface **603** thereof, visual depiction **622** rendered from another set of received customer inputs shows better uniformity in portion **624** of the visual depiction **622** indicative of a lesser need for the product being offered for sale in terms of the individual product need assessment categories of the number of transponders **614-1**, the discounts **614-2**, the managing system **614-7**, and the time allocated **614-8**. Conversely, portion **626** of the visual depiction **622** is indicative of a greater need for the product being offered for sale in terms of the individual product need assessment categories of the violations **614-3**, the consolidation **614-4**, the stress level **614-5**, and the need to manage **614-6**. Further, as shown in FIG. 9 and graphical interface **605** thereof, the visual depiction **628** rendered from yet another set of received consumer inputs shows the desired (i.e., best) uniformity **630** and is indicative of a lesser (or no) need for the product being offered for sale in terms of each of the individual product need assessment categories. Further, in accordance with an embodiment, in addition to informing the consumer in a real-time, interactive way with respect to a particular product need, the graphical interface and rendered visual depiction (e.g., graphical interface **605** and the visual depiction **628**) also assist the seller (e.g., seller 1 **118-1**) in their ability to adapt their sales funnel approach and/or product offering to address inefficiencies in respect to their overall product sales approach including, but not limited to, the performance of individual sales associates that are representing the company and selling their products.

(28) Turning our attention to FIG. 5, a flowchart of illustrative operations **500** are shown for providing an interactive sales funnel management tool for use with the interactive sales funnel system of FIG. 2 in accordance with an embodiment. In accordance with FIG. 5, the operations **500** comprising, at step **502**, generating, by the processor, a graphical interface for display on the display device. As detailed above, the graphical interface comprising a product need visualization wheel together with a visual questionnaire, and the product need visualization wheel comprising a form of a circular wheel with a center point, a circumference, and a plurality of spokes, each spoke configured between the center point and a particular one point along the circumference of the wheel in a symmetric configuration thereabout. Further, the visual questionnaire comprising a plurality of individual product need assessment categories specific to a product offered for sale to a consumer in accordance with criteria associated with a sales funnel, wherein each individual product need assessment category of the plurality of product need assessment categories having a corresponding graduated consumer assessment scale associated therewith. As detailed above, the graduated consumer assessment scale comprising a plurality of metrics for establishing the consumer's need for the product offered for sale, and each particular one spoke of the plurality of spokes corresponding with a particular one individual product need assessment category of the plurality of product need assessment categories. In this way, the product need visualization wheel

showing, along a respective one spoke of the plurality of spokes, a respective one graduated consumer assessment scale associated with each individual product need assessment category of the plurality of individual product need assessment categories.

(29) Then, at step **504**, displaying, on the display device, the graphical interface generated (see, e.g., FIGS. **6-9**) and at step **506** receiving, from the consumer, a plurality of inputs as a function of the plurality of individual product need assessment categories displayed by the graphical interface and the corresponding graduated consumer assessment scale associated therewith. As detailed herein above, each input of the plurality of inputs received being specific to a respective one individual product need assessment category of the plurality of individual product need assessment categories and comprising at least a selection of a respective one metric of the plurality of metrics from the graduated consumer assessment scale. At step **508**, updating, by the processor, the graphical interface displayed as function of the plurality of inputs received such that the updated graphical interface shows a respective one graduated consumer assessment scale associated with each individual category of a plurality of individual categories together with each input of the plurality of inputs received and the respective one metric of the plurality of metrics from the graduated consumer assessment scale selected, and generating, as part of the graphical interface updated, a visual depiction of a relationship between each respective one input received in relation to each other.

(30) In this way, the operations facilitate an interactive engagement, by and through a specifically configured application executing, for example, on the interactive sales funnel system **200**, with the individual consumer that allows the consumer to play an important role in the overall sales process and coming to their own realization of whether they need a particular product or not based on a customized needs assessment. That is, the customer is actively engaged and formulates their own product need assessment, for example, through the AIA model steps of (1) awareness—the customer is aware of the existence of a product or service; (2) interest—actively expressing an interest in a product group; (3) desire—aspiring to a particular brand or product; and (4) action—taking the next step towards purchasing the chosen product. This interactive engagement, in accordance with the disclosed embodiments, is facilitated directly by the customized visualization tool and graphical interface as generated, presented and updated in accordance with the principles of the disclosed embodiments herein.

(31) Continuing at step **510**, displaying the updated graphical interface updated; and at step **512** receiving from the consumer a selection, as a function of the graphical interface updated and the visual depiction of the relationship between each respective one input received in relation to each other, for the product offered for sale. As noted above, if the desired rendered relationship in any one visual depiction is that of a circle (see, e.g., FIG. **9**) then this relationship between the plurality of metrics is the most desirable and indicative of a lesser (or no) need for the product being offered for sale. Conversely, the further the displayed rendered relationship represents such circular shape (see, e.g., FIGS. **6-8**) this is an indication to the consumer of an identified need(s) for the product offered for sale and supports a decision to purchase such product. By navigating through the visual questionnaire and rendered product need visualization wheel(s), by and through the displayed graphical interface, the consumer can easily ascertain and visualize their need for a particular product as a function of their own direct product need assessments and not unduly influenced by others (e.g., a salesperson).

(32) As noted above, in some embodiments the method or methods described above may be executed or carried out by a computing system including a tangible computer-readable storage medium, also described herein as a storage machine, that holds machine-readable instructions executable by a logic machine (i.e., a processor or programmable control device) to provide, implement, perform, and/or enact the above-described methods, processes and/or tasks. When such methods and processes are implemented, the state of the storage machine may be changed to hold different data. For example, the storage machine may include memory devices such as various hard

disk drives, CD, or DVD devices. The logic machine may execute machine-readable instructions via one or more physical information and/or logic processing devices. For example, the logic machine may be configured to execute instructions to perform tasks for a computer program. The logic machine may include one or more processors to execute the machine-readable instructions. The computing system may include a display subsystem to display a graphical user interface (GUI), or any visual element of the methods or processes described above. For example, the display subsystem, storage machine, and logic machine may be integrated such that the above method may be executed while visual elements of the disclosed system and/or method are displayed on a display screen for user consumption. The computing system may include an input subsystem that receives user input. The input subsystem may be configured to connect to and receive input from devices such as a mouse, keyboard, or gaming controller. For example, a user input may indicate a request that certain task is to be executed by the computing system, such as requesting the computing system to display any of the above-described information or requesting that the user input updates or modifies existing stored information for processing. A communication subsystem may allow the methods described above to be executed or provided over a computer network. For example, the communication subsystem may be configured to enable the computing system to communicate with a plurality of personal computing devices. The communication subsystem may include wired and/or wireless communication devices to facilitate networked communication. The described methods or processes may be executed, provided, or implemented for a user or one or more computing devices via a computer-program product such as via an application programming interface (API).

(33) Thus, the steps of the disclosed method (see, e.g., FIG. 5) and the associated discussion herein above can be defined by the computer program instructions stored in a memory and/or data storage device and controlled by a processor executing the computer program instructions. Accordingly, by executing the computer program instructions, the processor executes an algorithm defined by the disclosed method. For example, the computer program instructions can be implemented as computer executable code programmed by one skilled in the art to perform the illustrative operations defined by the disclosed methods. Further, it will be appreciated that any flowcharts, flow diagrams, state transition diagrams, pseudo code, program code and the like represent various processes which may be substantially represented in computer readable medium and so executed by a computer, machine, or processor, whether or not such computer, machine or processor is explicitly shown. One skilled in the art will recognize that an implementation of an actual computer or computer system may have other structures and may contain other components as well, and that a high-level representation of some of the components of such a computer is for illustrative purposes.

(34) Since many modifications, variations, and changes in detail can be made to the described preferred embodiments of the invention, it is intended that all matters in the foregoing description and shown in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Thus, the scope of the invention should be determined by the appended claims and their legal equivalents.

Claims

1. An interactive sales funnel system comprising: a display device; a processor; a memory storing instructions that when executed cause the processor to perform operations comprising: generating, by the processor, a graphical interface for display on the display device, the graphical interface comprising a product need visualization wheel together with a visual questionnaire, the product need visualization wheel comprising a form of a circular wheel with a center point, a circumference and a plurality of spokes, each spoke configured between the center point and a particular one point along the circumference of the wheel in a symmetric configuration thereabout, the visual

questionnaire comprising a plurality of individual product need assessment categories specific to a product offered for sale to a consumer in accordance with criteria associated with a sales funnel, each individual product need assessment category of the plurality of product need assessment categories having a corresponding graduated consumer assessment scale associated therewith, the corresponding graduated consumer assessment scale comprising a plurality of metrics for establishing the consumer's need for the product offered for sale, each particular one spoke of the plurality of spokes corresponding with a particular one individual product need assessment category of the plurality of product need assessment categories, and the product need visualization wheel showing, along a respective one spoke of the plurality of spokes, a respective one graduated consumer assessment scale associated with each individual product need assessment category of the plurality of individual product need assessment categories such that the respective one graduated consumer assessment scale associated with each individual product need assessment category has a predetermined position along the circumference of the wheel; displaying, on the display device, the graphical interface generated, wherein each one of the respective one graduated consumer assessment scales associated with each individual product need assessment category is displayed at a predetermined position along the circumference of the wheel; receiving, from the consumer, a plurality of inputs as a function of the plurality of individual product need assessment categories displayed by the graphical interface and the corresponding graduated consumer assessment scale associated therewith, each input of the plurality of inputs received being specific to a respective one individual product need assessment category of the plurality of individual product need assessment categories and comprising at least a selection of a respective one metric of the plurality of metrics from the corresponding graduated consumer assessment scale; updating, by the processor, the graphical interface displayed as function of the plurality of inputs received such that the graphical interface updated shows a respective one graduated consumer assessment scale associated with each individual category of a plurality of individual categories together with each input of the plurality of inputs received and the respective one metric of the plurality of metrics from the corresponding graduated consumer assessment scale selected, and generating, as part of the graphical interface updated, a visual depiction of a relationship between each respective one input received in relation to each other; displaying, on the display device, the graphical interface updated, wherein each one of the respective one graduated consumer assessment scales associated with each individual product need assessment category is displayed by the graphical interface updated at the predetermined position along the circumference of the wheel thereby contributing to rendering of the visual depiction of the relationship between each respective one input received in relation to each other; and receiving, from the consumer a selection, as a function of the graphical interface updated and the visual depiction generated of the relationship between each respective one input received in relation to each other, for the product offered for sale.

2. The interactive sales funnel system of claim 1, wherein the sales funnel and the product offered for sale are directed to a toll management product that may be used by transportation professionals to manage infrastructure tolls collected by tolling authorities.
3. The interactive sales funnel system of claim 1, wherein the plurality of product need assessment categories comprise a number of transponders, discounts, violations, consolidation, stress level, need to manage, managing system and time allocated.
4. The interactive sales funnel system of claim 1, wherein the corresponding graduated consumer assessment scale is a numerical scale.
5. The interactive sales funnel system of claim 4, wherein the numerical scale comprises a range from 1 to 10, wherein an assessment of "1" represents a worst or lowest rating possible rating and an assessment of "10" represents a best or highest rating possible.
6. The interactive sales funnel system of claim 1, wherein a uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other is indicative of a need for the product for sale based on an assessment of the consumer.

7. The interactive sales funnel system of claim 6, wherein the greater the uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other represents a lesser need for the product offered for sale by the consumer.
8. The interactive sales funnel system of claim 6, wherein if the visual depiction generated of the relationship between each respective one input received in relation to each other in any one visual depiction is that of a circle, then the relationship between the plurality of metrics, as depicted by the visual depiction generated, is indicative of a lesser need for the product offered for sale by the consumer.
9. The interactive sales funnel system of claim 6, wherein the lesser the uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other represents a greater need for the product offered for sale by the consumer.
10. The interactive sales funnel system of claim 1, wherein a uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other is used to determine the consumer's need for at least the product offered for sale.
11. The interactive sales funnel system of claim 1, wherein the product need visualization wheel and the visual questionnaire are fully customizable.
12. The interactive sales funnel system of claim 11, wherein the customization of the product need visual wheel and the visual questionnaire is a function of a particular product and a particular market.
13. The interactive sales funnel system of claim 1, wherein the display device is part of a mobile device.
14. The interactive sales funnel system of claim 13, wherein the mobile device is one of a smartphone, a laptop computer, a tablet, and a wearable device.
15. An interactive sales funnel system comprising: a processor; a display device; a memory storing instructions that when executed cause the processor to perform operations comprising: generating, by the processor, a graphical interface for display on the display device, the graphical interface comprising a product need visualization wheel together with a visual questionnaire, the product need visualization wheel comprising a form of a circular wheel with a center point, a circumference and a plurality of spokes, each spoke configured between the center point and a particular one point along the circumference of the wheel in a symmetric configuration thereabout, the visual questionnaire comprising a plurality of individual product need assessment categories specific to a product offered for sale to a consumer in accordance with criteria associated with a sales funnel, each individual product need assessment category of the plurality of product need assessment categories having a corresponding graduated consumer assessment scale associated therewith, the corresponding graduated consumer assessment scale comprising a plurality of metrics for establishing the consumer's need for the product offered for sale, each particular one spoke of the plurality of spokes corresponding with a particular one individual product need assessment category of the plurality of product need assessment categories, and the product need visualization wheel showing, along a respective one spoke of the plurality of spokes, a respective one graduated consumer assessment scale associated with each individual product need assessment category of the plurality of individual product need assessment categories such that the respective one graduated consumer assessment scale associated with each individual product need assessment category has a predetermined position along the circumference of the wheel, and wherein the plurality of product need assessment categories comprise a number of transponders, discounts, violations, consolidation, stress level, need to manage, managing system and time allocated, and the sales funnel and the product offered for sale are directed to a toll management product that may be used by transportation professionals to manage infrastructure tolls collected by tolling authorities; displaying, on the display device, the graphical interface generated; receiving, from the consumer, a plurality of inputs as a function of the plurality of individual product need assessment categories displayed by the graphical interface and the corresponding graduated consumer

assessment scale associated therewith, each input of the plurality of inputs received being specific to a respective one individual product need assessment category of the plurality of individual product need assessment categories and comprising at least a selection of a respective one metric of the plurality of metrics from the corresponding graduated consumer assessment scale; updating, by the processor, the graphical interface displayed as function of the plurality of inputs received such that the graphical interface updated shows a respective one graduated consumer assessment scale associated with each individual category of a plurality of individual categories together with each input of the plurality of inputs received and the respective one metric of the plurality of metrics from the corresponding graduated consumer assessment scale selected, and generating, as part of the graphical interface updated, a visual depiction of a relationship between each respective one input received in relation to each other, and wherein a uniformity of the visual depiction of the relationship between each respective one input received in relation to each other is used to determine the consumer's need for at least the product offered for sale; displaying, on the display device, the graphical interface updated, wherein each one of the respective one graduated consumer assessment scales associated with each individual product need assessment category is displayed by the graphical interface updated at the predetermined position along the circumference of the wheel thereby contributing to rendering of the visual depiction of the relationship between each respective one input received in relation to each other; and receiving, from the consumer a selection, as a function of the graphical interface updated and the visual depiction generated of the relationship between each respective one input received in relation to each other, for the product offered for sale.

16. The interactive sales funnel system of claim 15, wherein the product need visualization wheel and the visual questionnaire are fully customizable as a function of a particular product and a particular market.

17. The interactive sales funnel system of claim 15, wherein the graduated consumer assessment scale is a numerical scale comprising a range from 1 to 10, wherein an assessment of “1” represents a worst or lowest rating possible rating and an assessment of “10” represents a best or highest rating possible.

18. The interactive sales funnel system of claim 15, wherein the greater the uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other represents a lesser need for the product offered for sale by the consumer.

19. The interactive sales funnel system of claim 15, wherein the lesser the uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other represents a greater need for the product offered for sale by the consumer.

20. An interactive sales funnel system for electronically signing an electronic document, the interactive sales funnel system comprising: a processor; a display device; a memory storing instructions that when executed cause the processor to perform operations comprising: generating, by the processor, a graphical interface for display on the display device, the graphical interface comprising a product need visualization wheel together with a visual questionnaire, the product need visualization wheel comprising a form of a circular wheel with a center point, a circumference and a plurality of spokes, each spoke configured between the center point and a particular one point along the circumference of the wheel in a symmetric configuration thereabout, the visual questionnaire comprising a plurality of individual product need assessment categories specific to a product offered for sale to a consumer in accordance with criteria associated with a sales funnel, each individual product need assessment category of the plurality of product need assessment categories having a corresponding graduated consumer assessment scale associated therewith, the corresponding graduated consumer assessment scale comprising a plurality of metrics for establishing the consumer's need for the product offered for sale, each particular one spoke of the plurality of spokes corresponding with a particular one individual product need assessment category of the plurality of product need assessment categories, and the product need visualization

wheel showing, along a respective one spoke of the plurality of spokes, a respective one graduated consumer assessment scale associated with each individual product need assessment category of the plurality of individual product need assessment categories such that the respective one graduated consumer assessment scale associated with each individual product need assessment category has a predetermined position along the circumference of the wheel, and wherein the plurality of product need assessment categories comprise a number of transponders, discounts, violations, consolidation, stress level, need to manage, managing system and time allocated, and the graduated consumer assessment scale is a numerical scale comprising a range from 1 to 10, wherein an assessment of “1” represents a worst or lowest rating possible rating and an assessment of “10” represents a best or highest rating possible; displaying, on the display device, the graphical interface generated; receiving, from the consumer, a plurality of inputs as a function of the plurality of individual product need assessment categories displayed by the graphical interface and the corresponding graduated consumer assessment scale associated therewith, each input of the plurality of inputs received being specific to a respective one individual product need assessment category of the plurality of individual product need assessment categories and comprising at least a selection of a respective one metric of the plurality of metrics from the graduated consumer assessment scale; updating, by the processor, the graphical interface displayed as function of the plurality of inputs received such that the graphical interface updated shows a respective one graduated consumer assessment scale associated with each individual category of a plurality of individual categories together with each input of the plurality of inputs received and the respective one metric of the plurality of metrics from the corresponding graduated consumer assessment scale selected, and generating, as part of the graphical interface updated, a visual depiction of a relationship between each respective one input received in relation to each other, and wherein a uniformity of the visual depiction of the relationship between each respective one input received in relation to each other is used to determine the consumer's need for at least the product offered for sale; displaying, on the display device, the graphical interface updated, wherein each one of the respective one graduated consumer assessment scales associated with each individual product need assessment category is displayed by the graphical interface updated at the predetermined position along the circumference of the wheel thereby contributing to rendering of the visual depiction of the relationship between each respective one input received in relation to each other; receiving, from the consumer a selection, as a function of the graphical interface updated and the visual depiction generated of the relationship between each respective one input received in relation to each other, for the product offered for sale; and wherein the greater the uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other represents a lesser need for the product offered for sale by the consumer, and the lesser the uniformity of the visual depiction generated of the relationship between each respective one input received in relation to each other represents a greater need for the product offered for sale by the consumer.
